9/April

• Matching Function (Theoretical Properties)

$$h = m(u(t), v(t))$$

h → # hires

m → matching function

$u(t)$ → unemployment

$v(t)$ → vaconcies: how many people firms want to hire

$m(u,v) \leq u$ (the matches cannot be greater than the # of pple looking for a job)

$m(u,v) \leq v$ (the matches cannot be greater than the # of pple the firms want to hire)

- unemployment exists ($u(t) \neq 0$), or if $u(t) = 0$ then $m(u=0, v) = 0$
  - ↳ counter cyclical
- vaconcy exists ($v(t) \neq 0$), or if $v(t) = 0$ then $m(u, v=0) = 0$
  - ↳ procyclical

- more unemployment gives you more matches

  $\frac{\partial m}{\partial u} > 0$
- more vaconcy gives you more matches

  $\frac{\partial m}{\partial v} > 0$

- Constant Returns to Scale (CRS)

  if $\lambda > 0$

  $m(\lambda u, \lambda v) = \lambda \cdot m(u, v)$

- **Tightness (θ)**: it's a ratio between buyers and sellers

$$\theta(t) = \frac{\text{buyers}}{\text{sellers}} : \frac{v(t)}{u(t)}$$

- Probability to find a job: $f(\theta) = \frac{h}{u} = \frac{m(u,v)}{u}$

$$= m\left(\frac{u}{u}, \frac{v}{u}\right)$$

inverse

$$\boxed{f(\theta) = m(1, \theta)}$$

- Probability to fill a vacancy: $q(\theta) = \frac{h}{v} = \frac{m(u,v)}{v}$

$$= m\left(\frac{u}{v}, \frac{v}{v}\right)$$

$$= m\left(\frac{1}{\theta}, 1\right)$$

$$\boxed{q(\theta) = m\left(\frac{1}{\theta}, 1\right)}$$

Labor market

- slack: $\uparrow u$, $\downarrow v$
- tight: $\downarrow u$, $\uparrow v$